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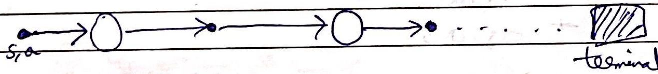
HW-3

~~Ans)~~ Initialize
$\pi(s) \in A(s)$ (arbitrarily) for all $s \in S$
$Q(s,a) \in R$ 0 ~~(arbitrarily)~~ for all $s \in S$, $a \in A(s)$
~~Mean(s,a) $\in R$ 0~~
Count(s,a) $\in R$ 0
Loop forever (for each episode):
Choose $S_0 \in S$, $A_0 \in A(S_0)$ all pair prob > 0
Generate episode
$G \leftarrow 0$
Loop for $t = T-1, T-2, \ldots, 0$
$G \leftarrow \gamma G + R_{t+1}$
Unless pair S_t, A_t in $S_0, A_0, \ldots, S_{t-1}, A_{t-1}$:
~~Append~~
$Q(s,a) \leftarrow \cancel{mean}\ Q(s,a) * count(s,a) + G$
$count(s,a) \leftarrow count(s,a) + 1$
$Q(s,a) \leftarrow Q(s,a) / count(s,a)$.
$\pi(S_t) \leftarrow \arg\max_a Q(s,a)$

Ans 2



Ans 3 $V(s) = \dfrac{\sum_{t \in T(s)} \rho_{t:T(t)-1} G_t}{\sum_{t \in T(s)} \rho_{t:T(t)-1}}$

~~We~~ We want T to track s,a instead of only s

$\therefore\ Q(s,a) = \dfrac{\sum_{t:T(s,a)} \rho_{t:(T(t)-1)} G_t}{\sum_{t \in T(s,a)} \rho_{t:(T(t)-1)}}$

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Ans 5. TD updates bootstrap while MC methods look at completely new episode. In case of new home example TD needs info about states from new home to highway the rest ~~so~~ states are known from bootstrapped knowledge. MC will sample new episodes and will be quite noisy in the beginning as it will fluctuate/adjust the highway states again. ~~Again~~

In the original scenario when learning for 1st time the TD might not always perform better than MC as both are still learning the path.

Ans 6. 6.3

The result shows that the episode terminated on the left block for 0 reward.

$$V(A) = V(A) + \alpha\left[R_{t+1} + V(S_T) - V(A)\right]$$

$$= 0.5 + 0.1\left[0 + 0 - 0.5\right]$$

$$= 0.45$$

So only $V(A)$ was updated. For all the rest, $V(S_t) = V(S) \Rightarrow V(S)$ ~~updat~~ too's not updated.

6.4

Long term accuracy of TD depends on chosen α. Smaller α leads to better convergence while larger α makes the value for oscillate. From the results TD performs better than MC as TD has more updates ~~due~~ as episode length is 6, TD will have 6 times more updates. Using wider α will have same behaviour. There is no fixed value of α where both algorithm would have performed better.

6.5

All states were initialized same. High α's exaggerate small errors causing the function to oscillate. Thus the error reduces but then increases again.

Ans 8 The update step of both algorithms will become almost similar, but the algorithms are still different. ~~SARSA~~

Consider both algorithms are currently at state S.
SARSA will use the ~~previous~~ greedy Action A from S to advance to ~~at~~ S' and then observe and update the value of $Q(S,a)$. Before updating ~~Q-learning~~ $Q(S,a)$ it will choose the A' from S'.

Q-learning

It will take greedy Action A from S to advance to S'. It will update $Q(S,a)$

Consider both algorithms at state $S_0 = S$.

SARSA

$S' = S$

Choose $A_0 = \arg\max_a Q_0(S,a)$
Take action A_0 and ~~get to~~ observe R, S'.
Choose Action $A_1 = \arg\max_a Q_0(S',a)$
Update ~~Evaluate~~ $Q_0(S,a) \rightarrow Q_1(S,a)$
~~Take A_1 and Observe R, S''~~

Q learning
~~A~~ choose $A_0 = \arg\max_a Q_0(S,a)$
Take A_0 and observe R, S'
Update ~~Evaluate~~ (S,a)
Choose $A_1 = \arg\max_a Q_1(S',a)$

If $s' = s$ then SARSA used the previous $Q_0(s,a)$ to take Action ~~update~~ while Q-learning will use updated $Q_1(s,a)$ to take action.

$\therefore$ weight update and Action selection will be different.